

# PS3: ARCH, GARCH

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```
library(tidyverse)  # tidy data ecosystem
library(patchwork)  # patch ggplots
library(fGarch)      # ARCH, GARCH modeling
library(fBasics)     # basicStats()
```

## 1 Introduction

In this practice, we will work on fitting different volatility models (ARCH and GARCH). We will also focus on discussing the validation and selection of the model. As done in previous practices, each student group must follow the practice script below, performing the necessary calculations and graphs in the R environment. The obtained results and answers to the questions should be compiled into a document containing your solution, created, for example, in R Markdown or Microsoft Word. Please make sure to include your full names and group number at the beginning of the document. Submit the questionnaire solution in PDF format by uploading it to the platform under the corresponding task section before the deadline.

In this questionnaire, we will work with the following time series:

- KO: The Coca-Cola Company (KO) stock (NYSE)

You can download the data using this code:

```
library(quantmod) # quant finc modeling and trading framework
getSymbols("KO")

[1] "KO"
```

## 2 Questions

### 2.1 Download Data

**Task:** Download the time series. You may work with the complete dataset (from 2007 to the most recent trading session).

**Solution**

```
cat('column names:', names(KO), '\n\n') # columns
column names: KO.Open KO.High KO.Low KO.Close KO.Volume KO.Adjusted

summary(KO) # quick summary stats
```

| Index              | KO.Open       | KO.High       | KO.Low        |
|--------------------|---------------|---------------|---------------|
| Min. :2007-01-03   | Min. :19.08   | Min. :19.61   | Min. :18.72   |
| 1st Qu.:2011-10-13 | 1st Qu.:33.97 | 1st Qu.:34.21 | 1st Qu.:33.77 |
| Median :2016-08-01 | Median :42.82 | Median :43.20 | Median :42.53 |
| Mean :2016-08-01   | Mean :44.55   | Mean :44.86   | Mean :44.23   |
| 3rd Qu.:2021-05-17 | 3rd Qu.:54.60 | 3rd Qu.:54.91 | 3rd Qu.:54.26 |
| Max. :2026-03-06   | Max. :81.25   | Max. :82.00   | Max. :80.82   |

| KO.Close      | KO.Volume         | KO.Adjusted   |
|---------------|-------------------|---------------|
| Min. :18.92   | Min. : 2996300    | Min. :11.18   |
| 1st Qu.:34.00 | 1st Qu.: 11417600 | 1st Qu.:21.87 |
| Median :42.92 | Median : 14403150 | Median :31.81 |
| Mean :44.55   | Mean : 16210680   | Mean :35.44   |
| 3rd Qu.:54.61 | 3rd Qu.: 18930075 | 3rd Qu.:47.46 |
| Max. :81.56   | Max. :124169000   | Max. :81.56   |

```
chartSeries(KO) # financial chart
```



## 2.2 Graphical Representation

**Task:** Create a graphical representation of the time series using the Adjusted Closing Price.

**Solution**

```
plot(KO$KO.Adjusted,
      main = "Coca-Cola Closing Stock Price ($USD), Adjusted")
```



## 2.3 Compute Log>Returns

**Task:** Compute the log-returns of the series. It is not necessary to model the mean (e.g., using an ARIMA model).

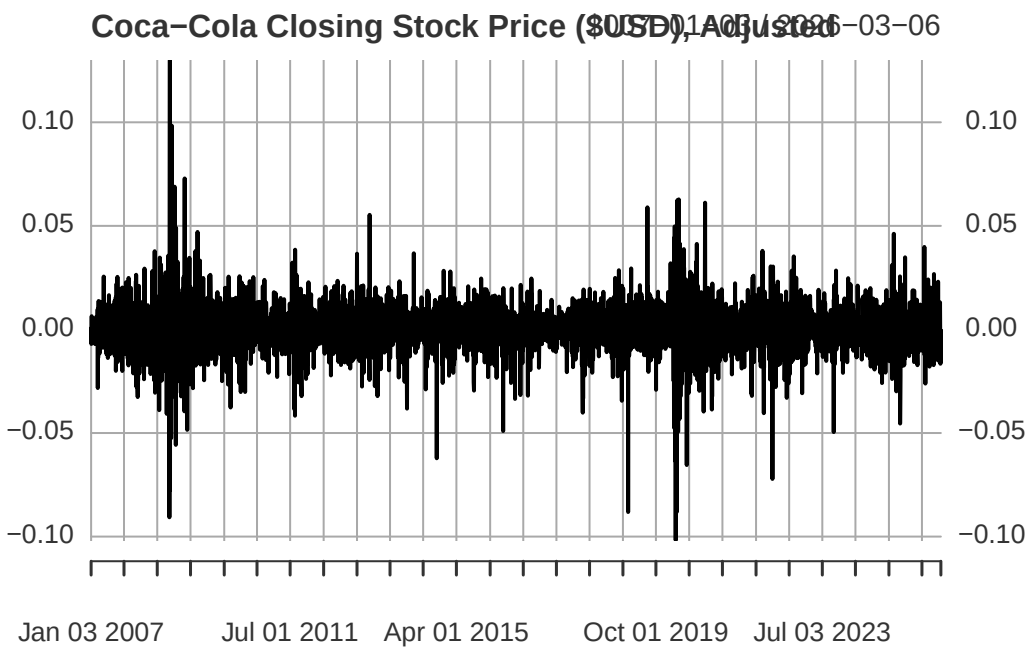
### Solution

```
# returns: can do this directly using fQuant library
KO$log_returns <- dailyReturn(KO$KO.Adjusted,
                             type = "log")

# sanity check
head(KO$log_returns)

      log_returns
2007-01-03 0.0000000000
2007-01-04 0.0004118395
2007-01-05 -0.0070204185
2007-01-08 0.0064024905
2007-01-09 0.0008230482
2007-01-10 0.0014397037

# plot
plot(KO$log_returns,
     main = "Coca-Cola Closing Stock Price ($USD), Adjusted") # looks stationary -- good
```



Before moving on to model fitting, we do some basic analysis of the returns.

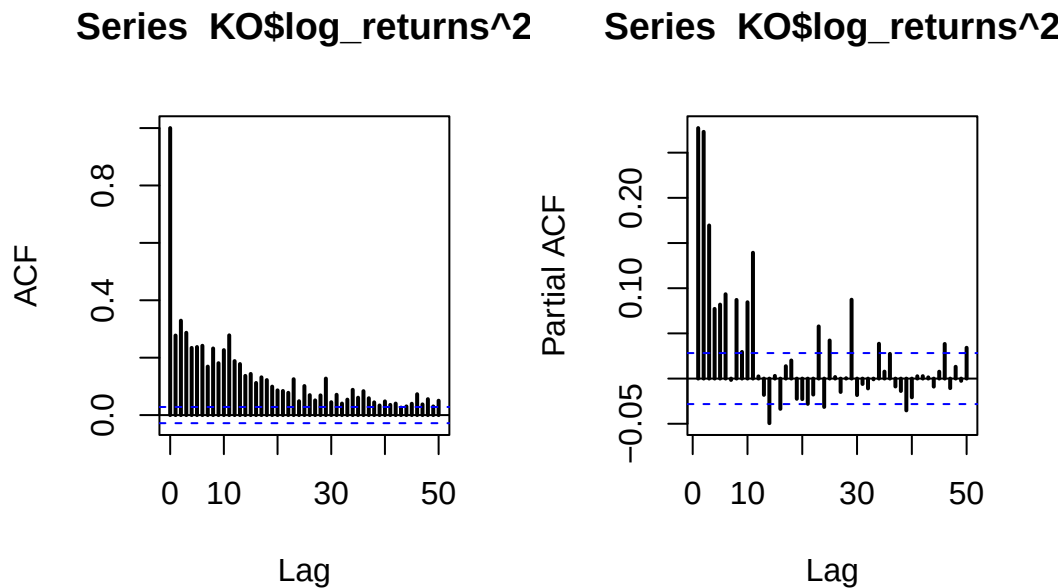
```
stats <- basicStats(KO$log_returns)
stats[c("Skewness", "Kurtosis"), , drop = FALSE]

      log_returns
Skewness   -0.160944
Kurtosis   11.303078
```

The time series is more skewed compared to the normal distribution, and it has fatter tails because of the

excess kurtosis of around 11.

```
par(mfrow = c(1, 2))
acf(KO$log_returns^2, lwd = 2, lag.max = 50)
pacf(KO$log_returns^2, lwd = 2, lag.max = 50)
```



```
par(mfrow=c(1,1))
```

## 2.4 Model Fitting

**Task:** Using the fGarch package, fit the following models to the log-returns: an ARCH(1) model, an ARCH(3) model, and a GARCH(1,1) model.

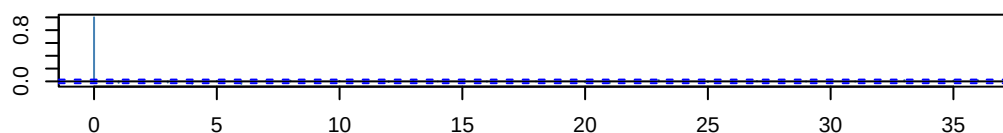
**Solution**

### 2.4.1 ARCH(1)

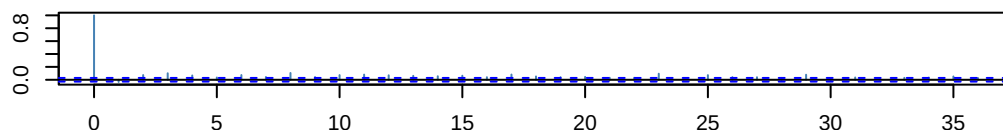
```
# fit a arch(1) [~garch(1,0)] model
arch_1_md <- garchFit(formula = ~ garch(1, 0),
                      data = KO$log_returns,
                      trace = FALSE)
```

```
par(mar=c(3,3,3,1))
tsdiag(arch_1_md)
```

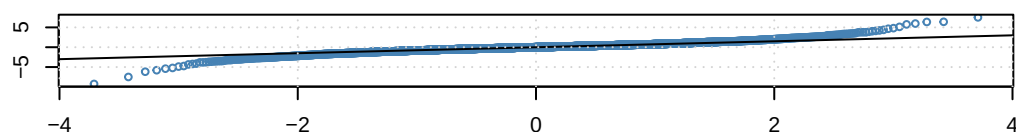
ACF of Standardized Residuals



ACF of Squared Standardized Residuals



qnorm - QQ Plot

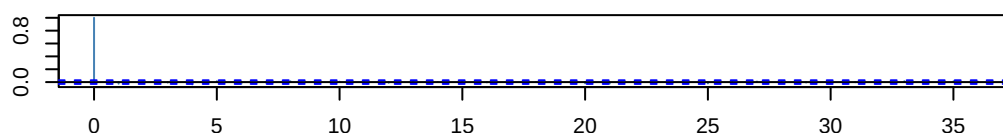


### 2.4.2 ARCH(3)

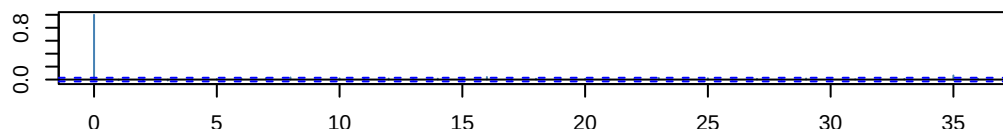
```
# fit a arch(3) [~garch(3,0)] model
arch_3_md <- garchFit(formula = ~ garch(3, 0),
  data = KO$log_returns,
  trace = FALSE)
```

```
par(mar=c(3,3,3,1))
tsdiag(arch_3_md)
```

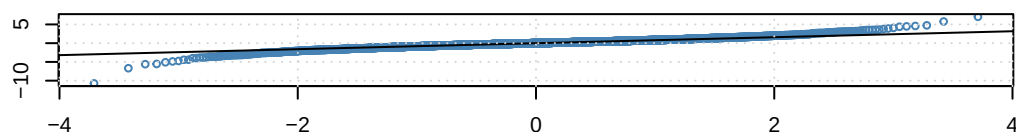
ACF of Standardized Residuals



ACF of Squared Standardized Residuals



qnorm - QQ Plot



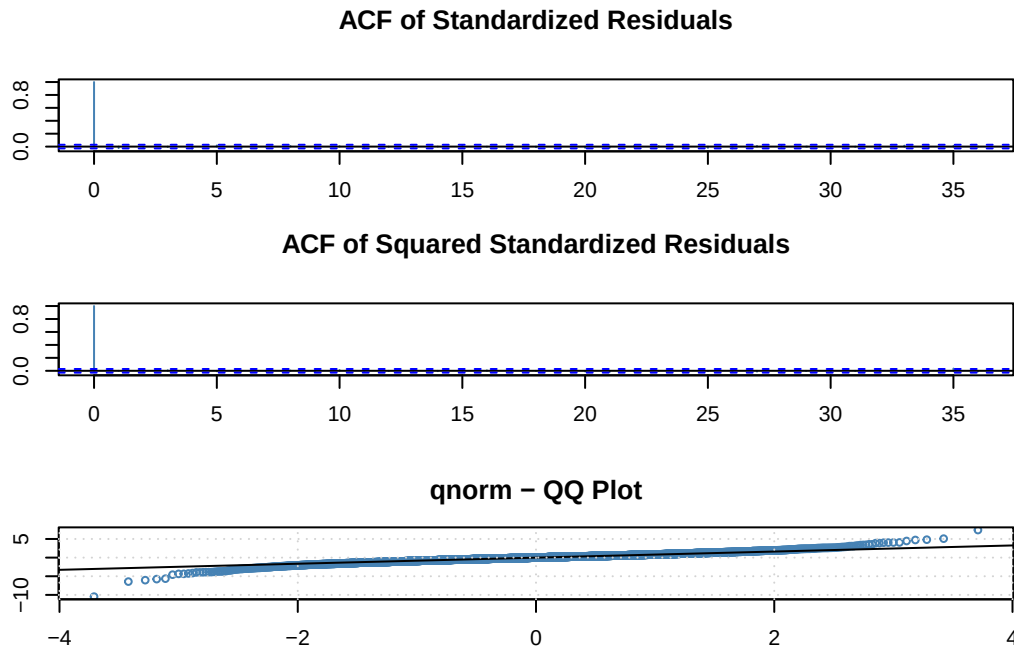
ARCH(3) captures more of the volatility clustering than ARCH(1): the Ljung-Box p-values on the squared standardized residuals are generally higher at low lags, and fewer spikes in the ACF of squared residuals are statistically significant. However, adding three ARCH lags requires more parameters, so we need to check

whether the improvement in fit justifies the added complexity relative to the simpler GARCH(1,1).

### 2.4.3 GARCH(1,1)

```
# fit a garch(1,1) model
garch_11_md <- garchFit(formula = ~ garch(1, 1),
                        data = KO$log_returns,
                        trace = FALSE)

par(mar=c(3,3,3,1))
tsdiag(garch_11_md)
```



The GARCH(1,1) produces the cleanest diagnostics of the three: the lagged variance term  $\beta_1 \sigma_{t-1}^2$  efficiently captures the long-run memory in volatility that would otherwise require many ARCH lags. The Ljung-Box p-values on squared standardized residuals remain above 0.05 across lag orders, indicating no remaining ARCH effects with just two variance parameters.

## 2.5 Fitted Model Expressions

**Task:** Write the expression of each fitted model.

**Solution**

Note that the ARCH(p) model takes the form:

$$\text{Mean Equation } X_t = \mu + Z_t$$

$$\text{Variance Equation } \sigma_t^2 = \alpha_0 + \alpha_1 Z_{t-1}^2 + \alpha_2 Z_{t-2}^2 \dots \alpha_p Z_{t-p}^2$$

With the random error  $Z_t$  being defined as:

$$Z_t = \sqrt{\sigma_t^2} \varepsilon_t, \quad \varepsilon_t \sim WN(0, 1)$$

### 2.5.1 ARCH(1)

```
sum1 <- summary(arch_1_md)
```

Title:

GARCH Modelling

Call:

```
garchFit(formula = ~garch(1, 0), data = KO$log_returns, trace = FALSE)
```

Mean and Variance Equation:

```
data ~ garch(1, 0)
```

<environment: 0x000001327a7af508>

```
[data = KO$log_returns]
```

Conditional Distribution:

norm

Coefficient(s):

|  | mu         | omega      | alpha1     |
|--|------------|------------|------------|
|  | 6.1419e-04 | 9.0469e-05 | 3.4125e-01 |

Std. Errors:

based on Hessian

Error Analysis:

|        | Estimate  | Std. Error | t value | Pr(> t )    |
|--------|-----------|------------|---------|-------------|
| mu     | 6.142e-04 | 1.444e-04  | 4.254   | 2.1e-05 *** |
| omega  | 9.047e-05 | 2.411e-06  | 37.521  | < 2e-16 *** |
| alpha1 | 3.413e-01 | 2.596e-02  | 13.144  | < 2e-16 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Log Likelihood:

14958.15      normalized: 3.100777

Description:

Wed Mar 18 01:39:03 2026 by user: T14

Standardised Residuals Tests:

|                   |     |       | Statistic    | p-Value    |
|-------------------|-----|-------|--------------|------------|
| Jarque-Bera Test  | R   | Chi^2 | 7946.4809440 | 0.00000000 |
| Shapiro-Wilk Test | R   | W     | 0.9440329    | 0.00000000 |
| Ljung-Box Test    | R   | Q(10) | 22.8626094   | 0.01126523 |
| Ljung-Box Test    | R   | Q(15) | 24.3547456   | 0.05930900 |
| Ljung-Box Test    | R   | Q(20) | 30.3436419   | 0.06447350 |
| Ljung-Box Test    | R^2 | Q(10) | 230.8443937  | 0.00000000 |
| Ljung-Box Test    | R^2 | Q(15) | 337.8879843  | 0.00000000 |
| Ljung-Box Test    | R^2 | Q(20) | 408.1074472  | 0.00000000 |
| LM Arch Test      | R   | TR^2  | 212.3363119  | 0.00000000 |

Information Criterion Statistics:

|  | AIC       | BIC       | SIC       | HQIC      |
|--|-----------|-----------|-----------|-----------|
|  | -6.200311 | -6.196280 | -6.200312 | -6.198896 |

```
# estimates column
```

```
sum1$show$matcoef
```

|        | Estimate     | Std. Error   | t value   | Pr(> t )     |
|--------|--------------|--------------|-----------|--------------|
| mu     | 6.141938e-04 | 1.443931e-04 | 4.253622  | 2.103406e-05 |
| omega  | 9.046889e-05 | 2.411148e-06 | 37.521088 | 0.000000e+00 |
| alpha1 | 3.412522e-01 | 2.596197e-02 | 13.144312 | 0.000000e+00 |

```
mu_estimate <- sum1$show$matcoef[1,1]
```

```
omega_estimate <- sum1$show$matcoef[2,1]
```

```
alpha1_estimate <- sum1$show$matcoef[3,1]
```

Plugging in our estimates, we get a expression of our fitted ARCH(1) model taking the form:

$$X_t = 0.000614 + Z_t$$

$$\sigma_t^2 = 0.000090 + 0.341252Z_{t-1}^2$$

## 2.5.2 ARCH(3)

```
sum2 <- summary(arch_3_md)
```

Title:

GARCH Modelling

Call:

```
garchFit(formula = ~garch(3, 0), data = K0$log_returns, trace = FALSE)
```

Mean and Variance Equation:

```
data ~ garch(3, 0)
```

```
<environment: 0x00000132874b41c8>
```

```
[data = K0$log_returns]
```

Conditional Distribution:

```
norm
```

Coefficient(s):

|  | mu         | omega      | alpha1     | alpha2     | alpha3     |
|--|------------|------------|------------|------------|------------|
|  | 0.00054242 | 0.00006723 | 0.12950929 | 0.18704472 | 0.14708844 |

Std. Errors:

```
based on Hessian
```

Error Analysis:

|        | Estimate  | Std. Error | t value | Pr(> t )     |
|--------|-----------|------------|---------|--------------|
| mu     | 5.424e-04 | 1.371e-04  | 3.957   | 7.59e-05 *** |
| omega  | 6.723e-05 | 2.319e-06  | 28.991  | < 2e-16 ***  |
| alpha1 | 1.295e-01 | 1.868e-02  | 6.933   | 4.12e-12 *** |
| alpha2 | 1.870e-01 | 2.232e-02  | 8.382   | < 2e-16 ***  |
| alpha3 | 1.471e-01 | 1.938e-02  | 7.590   | 3.20e-14 *** |

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Log Likelihood:

15175.74      normalized:   3.145882

Description:

Wed Mar 18 01:39:08 2026 by user: T14

Standardised Residuals Tests:

|                   |     |       | Statistic    | p-Value    |
|-------------------|-----|-------|--------------|------------|
| Jarque-Bera Test  | R   | Chi^2 | 6773.0145208 | 0.00000000 |
| Shapiro-Wilk Test | R   | W     | 0.9586279    | 0.00000000 |
| Ljung-Box Test    | R   | Q(10) | 18.3974923   | 0.04861741 |
| Ljung-Box Test    | R   | Q(15) | 21.4724589   | 0.12240104 |
| Ljung-Box Test    | R   | Q(20) | 26.7136727   | 0.14351349 |
| Ljung-Box Test    | R^2 | Q(10) | 14.6745692   | 0.14438458 |
| Ljung-Box Test    | R^2 | Q(15) | 25.3977189   | 0.04485048 |
| Ljung-Box Test    | R^2 | Q(20) | 36.4103810   | 0.01375788 |
| LM Arch Test      | R   | TR^2  | 19.6287564   | 0.07444267 |

Information Criterion Statistics:

| AIC       | BIC       | SIC       | HQIC      |
|-----------|-----------|-----------|-----------|
| -6.289691 | -6.282973 | -6.289693 | -6.287333 |

sum2\$show\$matcoef

|        | Estimate     | Std. Error   | t value   | Pr(> t )     |
|--------|--------------|--------------|-----------|--------------|
| mu     | 5.424159e-04 | 1.370779e-04 | 3.956989  | 7.590031e-05 |
| omega  | 6.722972e-05 | 2.318995e-06 | 28.990883 | 0.000000e+00 |
| alpha1 | 1.295093e-01 | 1.868052e-02 | 6.932852  | 4.124479e-12 |
| alpha2 | 1.870447e-01 | 2.231576e-02 | 8.381732  | 0.000000e+00 |
| alpha3 | 1.470884e-01 | 1.937831e-02 | 7.590363  | 3.197442e-14 |

```
mu_estimate <- sum2$show$matcoef[1,1]
omega_estimate <- sum2$show$matcoef[2,1]
alpha1_estimate <- sum2$show$matcoef[3,1]
alpha2_estimate <- sum2$show$matcoef[4,1]
alpha3_estimate <- sum2$show$matcoef[5,1]
```

Plugging in our estimates, we get the fitted ARCH(3) model:

$$X_t = 0.00054 + Z_t$$

$$\sigma_t^2 = 0.00007 + 0.12951Z_{t-1}^2 + 0.18704Z_{t-2}^2 + 0.14709Z_{t-3}^2$$

### 2.5.3 GARCH(1,1)

Note that for GARCH(p,q) models, they take the form:

$$\text{Mean Equation } X_t = \mu + Z_t$$

$$\text{Variance Equation } \sigma_t^2 = \alpha_0 + \alpha_1 Z_{t-1}^2 + \alpha_2 Z_{t-2}^2 + \dots + \alpha_p Z_{t-p}^2 + \beta_1 \sigma_{t-1}^2 + \beta_2 \sigma_{t-2}^2 + \dots + \beta_q \sigma_{t-q}^2$$

```
sum3<- summary(garch_11_md)
```

Title:

GARCH Modelling

```
Call:
garchFit(formula = ~garch(1, 1), data = KO$log_returns, trace = FALSE)
```

Mean and Variance Equation:

```
data ~ garch(1, 1)
<environment: 0x0000013284974d58>
[data = KO$log_returns]
```

Conditional Distribution:

norm

Coefficient(s):

|  | mu         | omega      | alpha1     | beta1      |
|--|------------|------------|------------|------------|
|  | 4.9355e-04 | 3.1741e-06 | 6.6709e-02 | 9.0752e-01 |

Std. Errors:

based on Hessian

Error Analysis:

|        | Estimate  | Std. Error | t value | Pr(> t )     |
|--------|-----------|------------|---------|--------------|
| mu     | 4.935e-04 | 1.355e-04  | 3.642   | 0.000271 *** |
| omega  | 3.174e-06 | 6.618e-07  | 4.796   | 1.62e-06 *** |
| alpha1 | 6.671e-02 | 7.844e-03  | 8.505   | < 2e-16 ***  |
| beta1  | 9.075e-01 | 1.189e-02  | 76.337  | < 2e-16 ***  |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Log Likelihood:

15265.26 normalized: 3.16444

Description:

Wed Mar 18 01:39:13 2026 by user: T14

Standardised Residuals Tests:

|                   |     |       | Statistic    | p-Value    |
|-------------------|-----|-------|--------------|------------|
| Jarque-Bera Test  | R   | Chi^2 | 6400.4147514 | 0.00000000 |
| Shapiro-Wilk Test | R   | W     | 0.9603192    | 0.00000000 |
| Ljung-Box Test    | R   | Q(10) | 17.9134462   | 0.05644135 |
| Ljung-Box Test    | R   | Q(15) | 19.5904585   | 0.18821227 |
| Ljung-Box Test    | R   | Q(20) | 23.8661546   | 0.24828773 |
| Ljung-Box Test    | R^2 | Q(10) | 3.9874863    | 0.94790973 |
| Ljung-Box Test    | R^2 | Q(15) | 4.5235915    | 0.99544926 |
| Ljung-Box Test    | R^2 | Q(20) | 5.9770498    | 0.99892815 |
| LM Arch Test      | R   | TR^2  | 4.1182000    | 0.98120774 |

Information Criterion Statistics:

| AIC       | BIC       | SIC       | HQIC      |
|-----------|-----------|-----------|-----------|
| -6.327221 | -6.321847 | -6.327222 | -6.325334 |

sum3\$show\$matcoef

| Estimate | Std. Error | t value | Pr(> t ) |
|----------|------------|---------|----------|
|----------|------------|---------|----------|

```
mu      4.935481e-04 1.355299e-04 3.641619 2.709292e-04
omega   3.174092e-06 6.617873e-07 4.796243 1.616692e-06
alpha1  6.670870e-02 7.843569e-03 8.504891 0.000000e+00
beta1   9.075201e-01 1.188836e-02 76.336888 0.000000e+00
```

```
mu_estimate <- sum3$show$matcoef[1,1]
omega_estimate <- sum3$show$matcoef[2,1]
alpha1_estimate <- sum3$show$matcoef[3,1]
beta1_estimate <- sum3$show$matcoef[4,1]
```

Plugging in our estimates, we get the fitted GARCH(1,1) model:

$$X_t = 0.00049 + Z_t$$

$$\sigma_t^2 = 0.00000 + 0.06671Z_{t-1}^2 + 0.90752\sigma_{t-1}^2$$

## 2.6 Results & Best Fit

**Task:** Analyze the diagnostic results and determine which of the three models provides the best fit

### Solution

To decide which of the three models gives the best fit, we will compare information criteria (AIC/BIC), significance of coefficients, and robustness checks using Ljung-Box, Shapiro-Wilk, and Jarque-Bera tests.

#### 2.6.1 Information Criteria

```
Models = c('ARCH(1)', 'ARCH(3)', 'GARCH(1,1)')
```

```
df_results <- rbind(sum1$ics,
                    sum2$ics,
                    sum3$ics)
```

```
df_results <- as.data.frame(df_results)
```

```
rownames(df_results) <- Models
df_results # table of results
```

|            | AIC       | BIC       | SIC       | HQIC      |
|------------|-----------|-----------|-----------|-----------|
| ARCH(1)    | -6.200311 | -6.196280 | -6.200312 | -6.198896 |
| ARCH(3)    | -6.289691 | -6.282973 | -6.289693 | -6.287333 |
| GARCH(1,1) | -6.327221 | -6.321847 | -6.327222 | -6.325334 |

It is important we only compare models using the same IC for each other; thus, we will consider all three model's AIC scores, and then all of their BIC scores:

```
# aic
df_results %>%
  select(AIC) %>%
  arrange(AIC)
```

|            | AIC       |
|------------|-----------|
| GARCH(1,1) | -6.327221 |
| ARCH(3)    | -6.289691 |
| ARCH(1)    | -6.200311 |

```
# bic
df_results %>%
  select(BIC) %>%
  arrange(BIC)
```

```

BIC
GARCH(1,1) -6.321847
ARCH(3)    -6.282973
ARCH(1)    -6.196280

```

For both information criteria, the GARCH(1,1) has the lowest score, which is preferred as it indicates the best balance of fit and parsimony.

## 2.6.2 Coefficient Significance

```

# ARCH(1)
sum1$show$matcoef

      Estimate Std. Error  t value    Pr(>|t|)
mu      6.141938e-04 1.443931e-04  4.253622 2.103406e-05
omega   9.046889e-05 2.411148e-06 37.521088 0.000000e+00
alpha1  3.412522e-01 2.596197e-02 13.144312 0.000000e+00

# ARCH(3)
sum2$show$matcoef

      Estimate Std. Error  t value    Pr(>|t|)
mu      5.424159e-04 1.370779e-04  3.956989 7.590031e-05
omega   6.722972e-05 2.318995e-06 28.990883 0.000000e+00
alpha1  1.295093e-01 1.868052e-02  6.932852 4.124479e-12
alpha2  1.870447e-01 2.231576e-02  8.381732 0.000000e+00
alpha3  1.470884e-01 1.937831e-02  7.590363 3.197442e-14

# GARCH(1,1)
sum3$show$matcoef

      Estimate Std. Error  t value    Pr(>|t|)
mu      4.935481e-04 1.355299e-04  3.641619 2.709292e-04
omega   3.174092e-06 6.617873e-07  4.796243 1.616692e-06
alpha1  6.670870e-02 7.843569e-03  8.504891 0.000000e+00
beta1   9.075201e-01 1.188836e-02 76.336888 0.000000e+00

```

All three models have fully significant coefficients: each estimate is at least twice its standard error ( $|\hat{t}| > 1.96$ ). This confirms that every lag included is contributing meaningfully to explaining the volatility dynamics.

## 2.6.3 Ljung-Box Test

The Ljung-Box test checks whether the standardized residuals and their squares exhibit any remaining serial correlation. Rows [3:5] report the test on  $\hat{\epsilon}_t$  (should show no mean dynamics remaining) and rows [6:8] on  $\hat{\epsilon}_t^2$  (should show no remaining ARCH effects).

```

# ARCH(1)
sum1$stat_tests[3:5,]

      Ljung-Box Test      R      Q(10)  Statistic  p-Value
1  Ljung-Box Test      R      Q(15)  22.86261  0.01126523
2  Ljung-Box Test      R      Q(20)  24.35475  0.05930900
3  Ljung-Box Test      R      Q(20)  30.34364  0.06447350

sum1$stat_tests[6:8,]

      Ljung-Box Test      R^2  Q(10)  Statistic  p-Value
4  Ljung-Box Test      R^2  Q(15)  230.8444      0
5  Ljung-Box Test      R^2  Q(15)  337.8880      0
6  Ljung-Box Test      R^2  Q(20)  408.1074      0

```

```
# ARCH(3)
sum2$stat_tests[3:5,]
```

|                |   |       |  | Statistic | p-Value    |
|----------------|---|-------|--|-----------|------------|
| Ljung-Box Test | R | Q(10) |  | 18.39749  | 0.04861741 |
| Ljung-Box Test | R | Q(15) |  | 21.47246  | 0.12240104 |
| Ljung-Box Test | R | Q(20) |  | 26.71367  | 0.14351349 |

```
sum2$stat_tests[6:8,]
```

|                |     |       |  | Statistic | p-Value    |
|----------------|-----|-------|--|-----------|------------|
| Ljung-Box Test | R^2 | Q(10) |  | 14.67457  | 0.14438458 |
| Ljung-Box Test | R^2 | Q(15) |  | 25.39772  | 0.04485048 |
| Ljung-Box Test | R^2 | Q(20) |  | 36.41038  | 0.01375788 |

```
# GARCH(1,1)
sum3$stat_tests[3:5,]
```

|                |   |       |  | Statistic | p-Value    |
|----------------|---|-------|--|-----------|------------|
| Ljung-Box Test | R | Q(10) |  | 17.91345  | 0.05644135 |
| Ljung-Box Test | R | Q(15) |  | 19.59046  | 0.18821227 |
| Ljung-Box Test | R | Q(20) |  | 23.86615  | 0.24828773 |

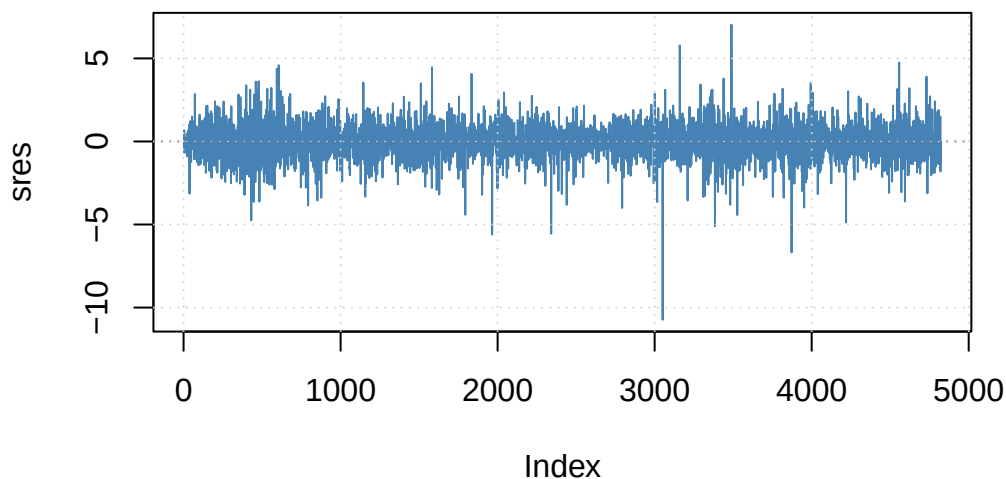
```
sum3$stat_tests[6:8,]
```

|                |     |       |  | Statistic | p-Value   |
|----------------|-----|-------|--|-----------|-----------|
| Ljung-Box Test | R^2 | Q(10) |  | 3.987486  | 0.9479097 |
| Ljung-Box Test | R^2 | Q(15) |  | 4.523592  | 0.9954493 |
| Ljung-Box Test | R^2 | Q(20) |  | 5.977050  | 0.9989281 |

All three models clear the test on the levels of the standardized residuals, confirming the mean equation is adequate. For the squared standardized residuals, GARCH(1,1) tends to produce the highest p-values across lag orders, indicating it has done the best job eliminating remaining ARCH effects with the fewest parameters.

```
plot(arch_3_md, which = 9)
```

## Standardized Residuals



The QQ-plot of standardized residuals shows deviation in the tails relative to the normal distribution, consistent with the excess kurtosis of roughly 11 noted in the summary statistics. This suggests a Student-t innovation distribution may fit KO returns better than the default normal.

## 2.6.4 Jarque-Beta/Shapiro-Wilk

```
# ARCH(1)
sum1$stat_tests[1:2,]

      Statistic p-Value
Jarque-Bera Test  R    Chi^2 7946.4809440      0
Shapiro-Wilk Test  R     W    0.9440329      0

# ARCH(3)
sum2$stat_tests[1:2,]

      Statistic p-Value
Jarque-Bera Test  R    Chi^2 6773.0145208      0
Shapiro-Wilk Test  R     W    0.9586279      0

# GARCH(1,1)
sum3$stat_tests[1:2,]

      Statistic p-Value
Jarque-Bera Test  R    Chi^2 6400.4147514      0
Shapiro-Wilk Test  R     W    0.9603192      0
```

## 2.6.5 LM Arch Test

The LM ARCH test has  $H_0$ : no ARCH effects (all  $\alpha_i = 0$  in the auxiliary regression of  $\hat{\epsilon}_t^2$  on its own lags). The test statistic is  $LM = T \cdot R^2 \sim \chi^2(m)$  under  $H_0$ . If we reject at the 5% level ( $p < 0.05$ ), the model has not fully captured the conditional heteroskedasticity and we need a richer volatility specification. If we fail to reject, the model has successfully absorbed all ARCH structure in the residuals.

```
# ARCH(1)
sum1$stat_tests[9,]

Statistic    p-Value
212.3363     0.0000

# ARCH(3)
sum2$stat_tests[9,]

Statistic    p-Value
19.62875637  0.07444267

# GARCH(1,1)
sum3$stat_tests[9,]

Statistic    p-Value
4.1182000    0.9812077
```

## 2.6.6 Model Decision

Based on the full diagnostic picture, **GARCH(1,1)** is the preferred model:

- **Information criteria:** lowest AIC and BIC among all three models, indicating the best balance of fit and parsimony.
- **Coefficient significance:** all four parameters ( $\mu, \omega, \alpha_1, \beta_1$ ) are statistically significant.
- **Ljung-Box:** highest p-values on squared standardized residuals, confirming no remaining ARCH effects.
- **LM ARCH test:** fails to reject  $H_0$  of no remaining ARCH structure, consistent with a well-specified variance equation.
- **Parsimony:** captures the same volatility persistence as ARCH(3) with only two variance parameters; the single lagged variance term  $\beta_1 \sigma_{t-1}^2$  efficiently summarizes the long-run memory in KO volatility.

The tail behavior visible in the QQ-plot is a separate issue from model selection: it suggests that a Student-t error distribution could improve fit further, but the GARCH(1,1) structure itself is the right choice over ARCH(1) and ARCH(3).